

Nickname:

1.8 Classwork: Standard deviation on the GDC

Use your GDC. Give answers exactly, or correct to three significant figures. Write down the values you put into the calculator.

- The maximum daily temperature, in degrees Celsius, was recorded in Rome on ten days in October.

Temperature (°C)	18	21	19	23	22	20	17	24	21	25
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- Find the mean and the standard deviation of these temperatures.
- Write down the variance of these temperatures.

A visitor from New York converts each temperature to degrees Fahrenheit using the formula $F = 1.8C + 32$.

- Find the mean of the ten temperatures in °F.
- Find the standard deviation of the ten temperatures in °F.
- Explain why the “+ 32” in the formula changes the mean but does not change the standard deviation.

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2. The table shows the number of goals, x , that a football team scored in each of its matches last season. The mean number of goals per match was 2.2.

Number of goals, x	0	1	2	3	4	5
Frequency, f	2	7	k	6	4	1

- Find the value of k .
- Find the standard deviation of the number of goals per match.
- Find the number of matches in which the number of goals was more than one standard deviation away from the mean.
- In its first match of the new season the team scores 2 goals. This match is added to the data. State whether the mean and the standard deviation increase, decrease or stay the same. Give a reason for each answer.